Pen Indicator States

MarketSys | Current state reference | October 25, 2026

STATE	SIGNAL	INDICATION
OFF		All colors off
ACTIVE		Solid green
LAST		Solid blue
HOLD_OPEN		Flashing green
FULL		Solid red
FAULT		Flashing red

Intended pen-routing flow

The penning app reports each head's destination after sale as it enters the penning alley. The controller derives the indicator changes.

Different destination: the new pen becomes ACTIVE; the previous ACTIVE pen becomes LAST; the older LAST pen becomes OFF unless it is the newly selected pen.

Same destination, another head: the active pen displays HOLD_OPEN. The LAST pen stays unchanged.

Retry of the same event: do not advance the flow or treat the retry as another head.

Current prototype behavior

Revision **0.4.2** is flashed and readback-verified on the controller and Nodes 1, 2 and 3. Flashing uses **500 ms on / 500 ms off**. Every non-OFF bench command automatically returns to OFF after **10 seconds**.

The indicators and addressed radio controls are implemented. Automatic pen-routing flow and the binary application interface remain planned. FULL and FAULT are available states; their automatic triggers remain to be defined.

Project references: Firmware deployment record | Pen-flow decisions